

CLOUD COMPUTING

Complete Study Notes — Day 1, Day 2 & Day 3

Topics Covered

Chapter 1 — What is Cloud?	Chapter 2 — Public vs Private Cloud
Chapter 3 — Virtualization	Chapter 4 — AWS vs Others
Chapter 5 — IAM in AWS	Chapter 6 — Practical: S3 + IAM Lab
Chapter 7 — EC2 in AWS	Chapter 8 — Practical: Jenkins on EC2
Chapter 9 — What is VPC + Components	Chapter 10 — How VPC Works + Why VPC
Chapter 11 — AWS Security Groups	Quick Revision — All Topics



CHAPTER 11 — AWS Security Groups



Simple Explanation — The Bouncer at a Club

Imagine you live in an apartment building.

The building has a main gate (that is your VPC)

Each flat has its own door lock (that lock = Security Group)

Your flat rules:

Family members can enter anytime → allowed

Pizza delivery can come to door → allowed on port 80

Strangers cannot enter → denied by default

You can go out anytime → outbound allowed

Every EC2, RDS, or Lambda in AWS has one of these locks.



Technical Definition



Definition:

A Security Group is a virtual STATEFUL firewall that controls inbound and outbound traffic at the resource level (instance level) in AWS.

It acts as the last line of defence before traffic reaches your actual server.

Term	Meaning
Stateful	If you allow inbound traffic, the response automatically goes out — no separate outbound rule needed
Inbound Rules	Controls what traffic is ALLOWED to come IN to your instance
Outbound Rules	Controls what traffic is ALLOWED to go OUT from your instance
Default Behaviour	Deny ALL inbound. Allow ALL outbound — always

Instance Level	Each EC2, RDS, Lambda gets its own Security Group attached
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Anatomy of a Security Group Rule — 5 Parts

Type	Protocol	Port	Source	Description
SSH	TCP	22	My IP only	Admin login
HTTP	TCP	80	0.0.0.0/0	Web traffic
HTTPS	TCP	443	0.0.0.0/0	Secure web
Custom TCP	TCP	8080	0.0.0.0/0	Jenkins
MySQL	TCP	3306	10.0.1.0/24	DB from app only

Most Important Ports to Know

Port	Used For
22 — SSH	Secure login to Linux servers (used in your Jenkins practical!)
80 — HTTP	Normal website traffic — any browser request
443 — HTTPS	Secure website traffic — encrypted
8080 — Custom TCP	Jenkins default port (you opened this in Chapter 8!)
3306 — MySQL	MySQL database connections
5432 — PostgreSQL	PostgreSQL database connections
3389 — RDP	Remote Desktop — login to Windows servers
27017 — MongoDB	MongoDB database connections

Stateful — The Most Important Concept

WITHOUT Stateful (NACL) User sends request on Port 80 → Inbound rule needed Response goes back to user → Outbound rule ALSO needed separately	WITH Stateful (Security Group) User sends request on Port 80 → Inbound rule needed Response goes back to user → AUTOMATIC — no extra rule needed
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Think of it like a phone call.

Once you pick up (allow inbound), you can talk back and forth freely.

You do not need a separate rule to speak back.

The Security Group REMEMBERS the connection and handles the response automatically.

Default Security Group Behaviour

Inbound Rules — Default

DENY ALL

Nothing gets in unless you explicitly add a rule to allow it.

Outbound Rules — Default

ALLOW ALL

Everything inside can freely reach the outside world.

Source Types When Writing Rules

Source Type	Example & Meaning
Anywhere IPv4	0.0.0.0/0 — Anyone on the internet can access
Anywhere IPv6	::/0 — Anyone on IPv6 internet
My IP	103.x.x.x/32 — Only YOUR specific IP address
Custom CIDR	10.0.1.0/24 — Only machines in that specific subnet
Another Security Group	sg-xxxxxxx — Only resources with that SG attached

 **BEST PRACTICE** — Never use 0.0.0.0/0 for SSH (Port 22)

If you open SSH to the world, bots will try to brute force your server within MINUTES. Always restrict SSH to your own IP only.

Security Group vs NACL — Full Comparison

Feature	Security Group	NACL
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Level	Instance level (EC2, RDS)	Subnet level (entire subnet)
State	Stateful	Stateless
Rule Types	Allow rules only	Allow AND Deny rules
Rule Evaluation	All rules at once	Top to bottom by number order
Default Inbound	Deny all	Allow all
Default Outbound	Allow all	Allow all
Think of as	Flat door lock	Floor security guard
When to use	Always — every resource	Extra subnet-level protection

Real World Setup — 3 Tier Architecture

TIER 1 — Web Server Security Group

Inbound: Port 80 from 0.0.0.0/0 (anyone can visit website)
 Inbound: Port 443 from 0.0.0.0/0 (anyone on HTTPS)
 Inbound: Port 22 from My IP only (only I can SSH)

TIER 2 — App Server Security Group

Inbound: Port 8080 from Web Server SG only (only web server talks to app)
 Inbound: Port 22 from My IP only

TIER 3 — Database Security Group

Inbound: Port 3306 from App Server SG ONLY (ONLY app server can query DB)

Result:

Internet to Web Server = Allowed
 Internet to App Server = Blocked
 Internet to Database = Blocked completely
 Web Server to Database = Blocked (cannot bypass app layer)

Security Group Best Practices

Best Practice	Why It Matters
Never open Port 22 to 0.0.0.0/0	Bots scan the internet for open SSH ports 24/7
Use separate SGs for each tier	Web, App, DB should each have their own rules
Use SG as source not IP ranges	More secure and auto-updates when instances change
Remove unused rules	Every open port is a potential attack surface

Name your SGs clearly	sg-web-server, sg-app-server, sg-database
Minimum access only	Only open what is absolutely needed — nothing more

Connection to Your Jenkins Practical

Remember in Chapter 8 — you added Port 8080 to the Security Group inbound rules.

Without that one rule, Jenkins was completely unreachable from the browser even though it was running perfectly inside the server.

That is exactly how Security Groups work in real life.
Running is not enough. The door must be opened.

 **Security Group = A smart stateful firewall attached to every AWS resource — it remembers connections, only allows what you explicitly permit, and blocks everything else by default.**



QUICK REVISION — All 4 Topics at a Glance

⚡ One-Liner Summaries

Topic	One Line Summary
☁ Cloud	Using someone else's powerful computer over the internet to store, run, or process YOUR data.
🌐 Public Cloud	Shared, cheap, managed by providers like AWS/Azure — great for startups and most businesses.
🔒 Private Cloud	Your own exclusive cloud — expensive but full control — used by banks, hospitals, governments.
💻 Virtualization	Dividing ONE physical machine into MULTIPLE virtual machines using a Hypervisor to save cost.
🏠 AWS	First, biggest, most trusted cloud — 200+ services, 400+ edge locations, 1M+ professionals.



Key Terms You Must Know

Term	Chapter	Meaning in Simple Words
Server	Cloud	A powerful computer that stores and serves your data
IaaS / PaaS / SaaS	Cloud	3 ways to buy cloud — rent hardware, platform, or software
Multi-tenant	Public Cloud	Many users share the same physical servers
Single-tenant	Private Cloud	Only one organization uses the infrastructure
On-premise	Private Cloud	Servers physically inside your own building
Hybrid Cloud	Both	Using both Public and Private Cloud together
Virtual Machine (VM)	Virtualization	A fake computer running inside a real one
Hypervisor	Virtualization	Software that creates and manages VMs
Snapshot	Virtualization	A saved state of a VM — like a game save point

Type 1 Hypervisor	Virtualization	Runs directly on hardware — used in data centers
Type 2 Hypervisor	Virtualization	Runs on top of OS — used on personal computers
EC2	AWS	Amazon's virtual server service
S3	AWS	Amazon's cloud storage service
Spot Instance	AWS	Cheapest AWS pricing — saves up to 90%
Edge Location	AWS	Mini data centers for fast content delivery (CDN)